THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 8: **Playing with Constructions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circle has centre O and radius 6 cm. Points are at distances 3, 6, 8, 5, 10 cm from O. How many of them lie inside the circle (not on it)? 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 4

2. A square has a diagonal that splits it into two triangles. If the side is 5 cm, the sum of the angles of one triangle is (degrees): 1

(a) 540 **(b)** 90 **(c)** 180 **(d)** 270

3. A rectangle's diagonals meet at O making $\angle AOB = 50^\circ$. Then $\angle AOD$ is (degrees): 1

(a) 140 **(b)** 160 **(c)** 130 **(d)** 120

4. A circle has diameter 24 cm. A concentric circle has radius 7 cm. The width of the ring (cm) is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

5. A square has perimeter 32 cm. Its diagonal splits it into two triangles each with two sides of length (cm): 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 24

6. Four equal circles of radius 3 cm are drawn touching in a row. The length of the row (cm) is: 1

(a) 36 **(b)** 24 **(c)** 27 **(d)** 21

7. A rectangle ABCD has AB = 14 cm. M is the midpoint of AB. AM (cm) is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 21

8. A circle has centre O and radius 9 cm. Points are at distances 9, 4, 12, 7, 2 cm from O. How many of them lie inside the circle (not on it)? 1

(a) 4 **(b)** 1 **(c)** 3 **(d)** 5

9. In rectangle ABCD, AB = 10 cm and BC = 7 cm. AC is a diagonal. The perimeter of rectangle (cm) is: 1

(a) 34 **(b)** 41 **(c)** 44 **(d)** 24

10. A rectangle's diagonals meet at O making $\angle AOB = 70^\circ$. Then $\angle AOD$ is (degrees): 1

(a) 100 **(b)** 120 **(c)** 140 **(d)** 110

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two concentric circles have radii 4 cm and 9 cm. Find the width of the ring between them, and the diameters of both circles. 2

12. In rectangle ABCD, $\angle AOB = 40^\circ$ at the point O where the diagonals meet. Find $\angle OAB$ and $\angle AOD$. 2

13. Find the area of a rectangle constructed with sides 10 cm and 7 cm and the area of a square of side 8 cm. Which is larger? 2

14. Two concentric circles have radii 5 cm and 12 cm. Find the width of the ring between them, and the diameters of both circles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A square of side 6 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

16. Three circles of radius 3 cm touch each other in a row. Find the distance between the centres of the first and third circles, and the total length of the row. 3

17. A rectangle has length 14 cm and its perimeter is 40 cm. Find its breadth and say whether it can be a square. 3

OR

A square of side 8 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

18. Three circles of radius 4 cm touch each other in a row. Find the distance between the centres of the first and third circles, and the total length of the row. 3

OR

A rectangle has length 15 cm and its perimeter is 60 cm. Find its breadth and say whether it can be a square. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the steps to construct a square ABCD of side 4 cm using only ruler and compass. If the full diagonal is 6 cm, state the half-diagonal OA and the angle formed by the diagonals. 5

OR

A gardener draws two concentric circles of radii 5 m and 8 m. Find the width of the path between them, and the diameters of the outer and inner circles. 5

20. Write the steps to construct a square ABCD of side 5 cm using only ruler and compass. If the full diagonal is 8 cm, state the half-diagonal OA and the angle formed by the diagonals. 5

OR

A gardener draws two concentric circles of radii 6 m and 10 m. Find the width of the path between them, and the diameters of the outer and inner circles. 5

END OF QUESTION PAPER — Total: 40 Marks